Global Well-Posedness, Neutral-State Rejection, and Ensemble-Induced Shell Coexistence in a Regularized Multicomponent Brazovskii Functional

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Abstract

We study one explicitly defined three-component complex field functional on a fixed periodic three-torus. The field is treated as six real components and the derivative Class-II terms are regularized by a positive density floor. Conditional only on the registered identification of this functional with the canonical P1 production functional, we prove global well-posedness of its real-L^2 gradient flow for every H^2 datum, exact energy dissipation, continuous dependence, and positive-time smoothness. An exact completion of the quadratic and local polynomial terms, together with a positive radial derivative of the Class-II energy, shows that the unconstrained functional has the zero field as its unique critical point and global minimizer and that the gradient flow decays exponentially in L^2. We then solve a distinct, imposed ensemble problem. On the side-16 torus, the exact spectral bottom lies on the integer shell $|n|^2 = 3$. After fixing $Q = \|\Psi\|_2^2/2$ or introducing $\mathcal{O}_\mu = \mathcal{F} - \mu Q$, an exact nonnegative decomposition gives a finite-shell constrained minimizer and a first-order grand-potential coexistence point. The charge or chemical potential is imposed rather than derived; the paper therefore makes no claim about a physical vacuum, BCC structure, infinite volume, or a quantum continuum limit. We provide a claim-to-proof map and a reproducibility protocol tied to the repository's independent exact audits.

1 Introduction and statement of scope

There are two mathematically different questions in the current TECT/A2 candidate. The first is whether a fixed regularized functional defines a well-behaved dissipative evolution and what its unconstrained equilibria are. The second is what happens after the variational problem is changed by fixing a nonzero quadratic charge or by subtracting a chemical-potential term. A nonzero minimizer of the second problem must not be reported as an equilibrium of the first problem. This distinction is the organizing principle of the present paper.

The object is a specific functional, not an assertion that this functional is the physical law of nature. Its identification with the canonical P1 production functional is retained as the named hypothesis `A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL`. All conclusions below are conditional on that identification and unconditional for the explicitly defined mathematical functional once the identification is accepted.

The paper has three main contributions.

(i) We give a continuum proof for the full regularized three-component fourth-order functional, including the derivative Class-II terms. The nonlinear map has order two, not four, and the endpoint H^4 estimate is obtained by a Duhamel cancellation rather than by an invalid fractional-smoothing shortcut.

(ii) We give an exact global sign test for the pinned unconstrained problem. It rejects every nonzero equilibrium of this candidate, not merely the finite list of lattice patterns tested previously.

(iii) We solve the separately declared fixed-charge and grand-canonical problems by an exact spectral and polynomial completion. The result is a finite-volume ensemble mechanism, with an explicit discontinuous charge jump, while the original neutral reference remains lower.

The statements are deliberately narrower than a constructive quantum-field theory theorem. We do not claim an infinite-volume limit, a real-time automorphism group, algebraic KMS dynamics, a ground-state gap, a continuum renormalization, a derived conserved charge, a density crystal, BCC selection, or any cosmological interpretation.

2 The functional and conventions

Let $\mathbb{T}^3_{16} = [0,16)^3$ with periodic boundary conditions and let $\Psi : \mathbb{T}^3_{16} \to \mathbb{C}^3$. We regard $\mathbb{C}^3$ as $\mathbb{R}^6$ and use the real pairing

$$\langle U, V\rangle_{\mathbb{R}} = \mathrm{Re}\int_{\mathbb{T}^3_{16}} U^\dagger V \, \mathrm{d}x. \tag{2.1}$$

Write $\rho = \Psi^\dagger\Psi$. For the three Hermitian internal generators T_A in the pinned P1 data, set

$$m_A = \Psi^\dagger T_A \Psi, \qquad J_A = \nabla m_A, \qquad K_A = J_A - \frac{m_A}{\rho+\varepsilon_\rho}\nabla\rho, \qquad \varepsilon_\rho = 10^{-12}. \tag{2.2}$$

The positive floor in (2.2) is part of the declared functional; removing it is a different theorem.

The quadratic operator is

$$L = (r + Z(-\Delta) + Y(-\Delta)^2)\mathrm{I}_3 + M_{\mathrm{fam}} + k_{\mathrm{lock}}(\mathrm{I}_3 - P_0), \tag{2.3}$$

where the pinned scalar coefficients are

$$Y = 1, \qquad Z = -0.9252754126, \qquad r = 0.4740336473, \tag{2.4}$$

and the family/lock contribution is positive semidefinite. The local polynomial coefficients are

$$\lambda = -\frac{43}{100}, \qquad \gamma = \frac{81}{50} > 0. \tag{2.5}$$

For the spectral calculation below, the sum $M_{\mathrm{fam}} + k_{\mathrm{lock}}(\mathrm{I} - P_0)$ is the explicit internal matrix displayed in (5.2). The Class-II density is

$$\mathcal{F}_{\mathrm{II}}(\Psi) = \sum_A \int_{\mathbb{T}^3_{16}} \left(\frac{a}{2}|J_A|^2 + b\,\mathrm{Re}(J_A^\dagger K_A) + \frac{c}{2}|K_A|^2\right) \mathrm{d}x, \tag{2.6}$$

with the pinned positive-definite coefficient matrix $Q_{\mathrm{II}} = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. The full functional is

$$\mathcal{F}(\Psi) = \frac{1}{2}\langle \Psi, L\Psi\rangle_{\mathbb{R}} + \int_{\mathbb{T}^3_{16}} \left(\frac{\lambda}{4}\rho^2 + \frac{\gamma}{6}\rho^3\right) \mathrm{d}x + \mathcal{F}_{\mathrm{II}}(\Psi). \tag{2.7}$$

The domain for the evolution theorem is $H^2(\mathbb{T}^3_{16}; \mathbb{C}^3)$, while the linear operator has domain H^4 and form domain H^2. All assertions about the charge ensemble below refer to the same fixed torus and the same functional (2.7); no new physical field or reservoir is introduced.

3 Main results

Theorem 3.1 (full gradient-flow well-posedness)**.** *Assume the named canonical-functional hypothesis H3. For every* $\Psi_0 \in H^2(\mathbb{T}^3_{16}; \mathbb{C}^3)$*, the real-*$L^2$ *gradient flow*

$$\partial_t \Psi = -\nabla_{\mathbb{R}} \mathcal{F}(\Psi), \qquad \Psi(0) = \Psi_0, \tag{3.1}$$

has a unique global solution with

$$\Psi \in C([0,\infty); H^2) \cap L^2_{\mathrm{loc}}((0,\infty); H^4), \qquad \partial_t \Psi \in L^2_{\mathrm{loc}}((0,\infty); L^2). \tag{3.2}$$

The solution depends continuously on the initial datum in H^2 *on every finite time interval and satisfies*

$$\mathcal{F}(\Psi(t)) + \int_s^t \|\partial_\tau \Psi(\tau)\|_2^2 \,\mathrm{d}\tau = \mathcal{F}(\Psi(s)), \qquad 0 \le s \le t. \tag{3.3}$$

For every $t > 0$ *it is smooth in space and time.*

Theorem 3.2 (neutral minimizer and exponential decay)**.** *For the pinned coefficients, there are exact rational constants*

$$g = \frac{7198187500025582338837}{54000000000000000000000} > \frac{1}{8}, \qquad \kappa = \frac{21016750000767470165 11}{81000000000000000000000} > \frac{1}{4} \tag{3.4}$$

such that every $\Psi \in H^2$ *obeys*

$$\mathcal{F}(\Psi) \ge g \,\|\Psi\|_2^2\,, \qquad \langle D\mathcal{F}(\Psi), \Psi\rangle_{\mathbb{R}} \ge \kappa \,\|\Psi\|_2^2\,. \tag{3.5}$$

Consequently $\Psi = 0$ *is the unique critical point and global minimizer of the unconstrained problem, and every solution of* (3.1) *obeys*

$$\|\Psi(t)\|_2^2 \le e^{-2\kappa t} \,\|\Psi_0\|_2^2\,. \tag{3.6}$$

Theorem 3.3 (imposed charge and chemical-potential ensemble)**.** *Let* $Q(\Psi) = \|\Psi\|_2^2/2$ *and let* λ_0 *be the exact bottom of the quadratic operator* (2.3) *on the side-*16 *torus. Put*

$$\rho_* = \frac{43}{216}, \qquad c_* = \frac{1849}{86400}, \qquad \mu_t = \lambda_0 - c_*. \tag{3.7}$$

Then, with $\rho = \Psi^\dagger \Psi$*,*

$$\mathcal{F}(\Psi) = \mu_t Q(\Psi) + \frac{1}{2}\langle \Psi, (L-\lambda_0)\Psi\rangle_{\mathbb{R}} + \mathcal{F}_{\mathrm{II}}(\Psi) + \frac{\gamma}{6}\int_{\mathbb{T}^3_{16}} \rho(\rho-\rho_*)^2 \,\mathrm{d}x. \tag{3.8}$$

Every term after $\mu_t Q$ *is nonnegative. A ground-shell plane wave with constant density* ρ_* *is a constrained global minimizer at*

$$Q_* = \frac{11008}{27}. \tag{3.9}$$

For $\mathcal{O}_\mu = \mathcal{F} - \mu Q$*, zero is the unique minimizer for* $\mu < \mu_t$*, coexists with nonzero ground-shell minimizers at* $\mu = \mu_t$*, and is beaten by a nonzero minimizer for* $\mu > \mu_t$*. The coexistence point lies strictly between the amplitude saddle-node and the zero-field spinodal:*

$$\mu_{\mathrm{sn}} = \lambda_0 - \frac{1849}{64800} < \mu_t = \lambda_0 - \frac{1849}{86400} < \lambda_0 = \mu_{\mathrm{lin}}. \tag{3.10}$$

This is a theorem about an imposed ensemble, not a derived physical charge or neutral-vacuum selection.

4 Proof of the evolution theorem

4.1 Linear operator and coercivity

The scalar symbol of (2.3) is $p(x) = r + Zx + Yx^2$ with $x = |k|^2$. Completing the square gives

$$p(x) = Y\left(x + \frac{Z}{2Y}\right)^2 + r - \frac{Z^2}{4Y}, \qquad \inf_{x\geq 0} p(x) = 0.260000000009475 > 0. \tag{4.1}$$

Moreover,

$$\inf_{x\geq 0} \frac{p(x)}{1+x^2} = 0.2048572626782363 > 0. \tag{4.2}$$

The family matrix and the lock term are positive semidefinite, so L is self-adjoint on H^4 and its form domain is H^2, with equivalent graph and Sobolev norms. The pinned Class-II matrix is positive definite; its recorded determinant and least eigenvalue are respectively

$$\det Q_{\mathrm{II}} = 7.031249999996483 \times 10^{-6}, \qquad \lambda_{\min}(Q_{\mathrm{II}}) = 0.001259011500926061. \tag{4.3}$$

Since $\lambda < 0$ and $\gamma > 0$, Young's inequality gives

$$\frac{|\lambda|}{4}\rho^2 \leq \frac{\gamma}{12}\rho^3 + C_Y, \qquad C_Y = 0.010098435197886498. \tag{4.4}$$

Combining (4.2), (4.3), and (4.4) yields, for a positive c_{H^2},

$$\mathcal{F}(\Psi) \geq \frac{c_{H^2}}{2}\|\Psi\|_{H^2}^2 + \frac{\gamma}{12}\|\Psi\|_{L^6}^6 - C_Y|\mathbb{T}_{16}^3|. \tag{4.5}$$

4.2 The Class-II map has order two

Realify Ψ as $u \in \mathbb{R}^6$. Let A_A be the real symmetric realification of T_A, and define

$$p_A(u) = 2A_A u, \qquad q_A(u) = \frac{u^T A_A u}{u^T u + \varepsilon_\rho}, \qquad s_A(u) = 2(A_A - q_A(u)\mathrm{I})u. \tag{4.6}$$

The Class-II density can then be written exactly as

$$G_{\mathrm{II}}(u, \nabla u) = \frac{1}{2}\sum_j (\partial_j u)^T B(u)(\partial_j u), \quad B(u) = \sum_A \{a p_A p_A^T + b(p_A s_A^T + s_A p_A^T) + c s_A s_A^T\}. \tag{4.7}$$

The positive ε_ρ makes B smooth on all of $\mathbb{R}^6$. Direct Euler–Lagrange differentiation therefore has the form

$$N_{\mathrm{II}}(u) = B(u)\nabla^2 u + C(u)[\nabla u, \nabla u], \tag{4.8}$$

where C is smooth and no derivative above order two occurs. To fix the notation used in the evolution equation, let the real gradient of the local polynomial be

$$N_{\mathrm{loc}}(u) = \lambda(u^T u)u + \gamma(u^T u)^2 u, \qquad N(u) = N_{\mathrm{loc}}(u) + N_{\mathrm{II}}(u). \tag{4.9}$$

The local term is order zero and is locally Lipschitz from H^2 to L^2; therefore the estimate below is for the full lower-order map N, not only its Class-II part. On the fixed three-torus, $H^2 \hookrightarrow L^\infty \cap W^{1,6}$ and $L^3 \hookrightarrow L^2$. Product estimates and the algebra property of H^2 thus give, on every bounded H^2 ball,

$$\|N(u) - N(v)\|_2 \leq C_R \|u - v\|_{H^2}. \tag{4.10}$$

The positive fourth-order operator generates an analytic semigroup [1, 2]. In particular,

$$\left\|L^{1/2} e^{-tL}\right\|_{L^2 \to L^2} \leq C t^{-1/2}, \qquad t > 0. \tag{4.11}$$

The kernel in (4.11) is integrable at the origin. The mild equation, (4.10), and a contraction argument in $C([0,T]; H^2)$ yield a unique local solution and the usual continuation alternative.

4.3 Galerkin passage and energy identity

Let P_n be the real L^2 Fourier projection and solve

$$\partial_t u_n = -P_n(Lu_n + N(u_n)). \tag{4.12}$$

This is the exact finite-dimensional real gradient flow, hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}(u_n) = -\left\|\partial_t u_n\right\|_2^2. \tag{4.13}$$

The coercivity estimate gives a uniform $L^\infty(0,T;H^2)$ bound and (4.13) gives an $L^2(0,T;L^2)$ bound on $\partial_t u_n$. Ellipticity applied to $Lu_n = -\partial_t u_n - P_n N(u_n)$ then gives an $L^2(0,T;H^4)$ bound. Aubin–Lions compactness [5, 6] yields strong convergence in $L^2(0,T;H^2)$, and (4.10) gives strong convergence of the nonlinear term in $L^2(0,T;L^2)$.

For the nonlinear part Φ, the derivative identity $D\Phi(u)[h] = \langle N(u), h\rangle_{\mathbb{R}}$ and the local Lipschitz estimate are understood in the Gelfand triple $H^2 \hookrightarrow L^2 \hookrightarrow H^{-2}$. In particular, $D\Phi : H^2 \to L^2 \subset H^{-2}$ is locally Lipschitz, while the solution belongs to $L^2(s,t;H^2) \cap H^1(s,t;H^{-2})$ on every finite interval. The Bochner chain rule therefore gives

$$\Phi(u(t)) - \Phi(u(s)) = \int_s^t \langle N(u(\tau)), \partial_\tau u(\tau)\rangle_{\mathbb{R}}\,\mathrm{d}\tau. \tag{4.14}$$

The quadratic part is $\frac{1}{2}\left\|L^{1/2}u\right\|_2^2$ and obeys the corresponding Hilbert-space chain rule under the same time regularity. Combining the two identities proves (3.3), not just an energy inequality. The lower bound (4.5) prevents the H^2 norm from diverging at a finite maximal time, so the solution is global.

4.4 Continuous dependence and smoothing

For two solutions in a common energy-bounded ball, the mild difference equation has the Volterra estimate

$$\left\|u(t) - v(t)\right\|_{H^2} \le C\left\|u_0 - v_0\right\|_{H^2} + C_R\int_0^t (t-s)^{-1/2}\left\|u(s) - v(s)\right\|_{H^2}\,\mathrm{d}s. \tag{4.15}$$

Weakly singular Gronwall proves continuous dependence on finite intervals. Analytic-semigroup estimates first give $u(t) \in D(L^\alpha) = H^{4\alpha}$ for $t > 0$ and every $\alpha < 1$. Choose $\alpha > 1/2$ to obtain Hölder continuity into H^2 away from $t = 0$. Thus $f(t) = N(u(t))$ is Hölder continuous into L^2, and the endpoint cancellation

$$L\int_a^t e^{-(t-s)L} f(s)\,\mathrm{d}s = (\mathrm{I} - e^{-(t-a)L})f(t) + \int_a^t Le^{-(t-s)L}[f(s) - f(t)]\,\mathrm{d}s \tag{4.16}$$

has an integrable last term. It follows that $u(t) \in D(L) = H^4$ for every $t > 0$. Periodic Moser estimates give

$$N : H^m \longrightarrow H^{m-2} \quad \text{locally Lipschitz,} \qquad m \ge 2, \tag{4.17}$$

and iteration of (4.16) proves the claimed C^∞ regularity.

5 Exact neutral-state result

5.1 Quadratic completion

The scalar symbol has the exact minimum

$$\mu_{\mathrm{sh}} = r - \frac{Z^2}{4Y} = \frac{26000000000947494031}{10^{20}} > 0. \tag{5.1}$$

The pinned internal matrix is

$$M=\begin{pmatrix}1/10 & -1/20 & -1/20\\ -1/20 & 13/100 & -1/20\\ -1/20 & -1/20 & 17/100\end{pmatrix}. \tag{5.2}$$

The three leading Sylvester minors of $M-(7/250)\mathrm{I}$ are

$$\frac{9}{125},\qquad \frac{1211}{250000},\qquad \frac{89}{31250000}, \tag{5.3}$$

so $M>(7/250)\mathrm{I}$. Parseval therefore gives

$$\mathcal{F}_{\text{quad}}(\Psi)\geq\frac{\nu}{2}\left\|\Psi\right\|_2^2,\qquad \nu=\mu_{\text{sh}}+\frac{7}{250}. \tag{5.4}$$

The full Class-II matrix is positive definite because, after writing $P=M_X^2+\varepsilon_X>0$, its determinant factor is

$$c_{JJ}c_{KK}-c_{JK}^2=\frac{1}{50}>0. \tag{5.5}$$

For the local polynomial, direct coefficient matching gives

$$\frac{\nu}{2}\rho+\frac{\lambda}{4}\rho^2+\frac{\gamma}{6}\rho^3=g\rho+\frac{\gamma}{6}\rho(\rho-\rho_*)^2,\quad \rho_*=-\frac{3\lambda}{4\gamma}=\frac{43}{216},\quad g=\frac{\nu}{2}-\frac{3\lambda^2}{32\gamma}. \tag{5.6}$$

The exact rational evaluation of g is the first value in (3.4). After integration and addition of the nonnegative Class-II term, (3.5)'s first inequality follows. Equality forces $\rho=0$ almost everywhere, so the zero field is the unique global minimizer.

5.2 Radial derivative and absence of nonzero equilibria

The energy lower bound alone does not rule out nonzero saddles. We therefore scale $\Psi\mapsto t\Psi$ and put $y=t^2$. At a fixed field, let j denote the unscaled current and let $\ell(y)=j-q(y)\nabla\rho$ with

$$q(y)=\frac{ym}{y\rho+\varepsilon_\rho},\qquad \theta=\frac{\varepsilon_\rho}{y\rho+\varepsilon_\rho}\in[0,1]. \tag{5.7}$$

Direct differentiation gives

$$\frac{\mathrm{d}E_{\text{II}}}{\mathrm{d}y}=y\begin{pmatrix}j & \ell\end{pmatrix}R_\theta\begin{pmatrix}j\\ \ell\end{pmatrix},\qquad R_\theta=\begin{pmatrix}a-\theta b & b+\theta(b-c)/2\\ b+\theta(b-c)/2 & c(1+\theta)\end{pmatrix}. \tag{5.8}$$

For the pinned coefficients,

$$a-\theta b\geq\frac{21}{2000P}>0,\qquad \det R_\theta=\frac{9(-81\theta^2+128\theta+128)}{10240000P^2}>0\quad(0\leq\theta\leq 1). \tag{5.9}$$

The numerator is concave and positive at both endpoints, so the Class-II energy is nondecreasing along every amplitude ray. Differentiating the remaining terms in the same ray gives

$$\langle D\mathcal{F}(\Psi),\Psi\rangle_{\mathbb{R}}\geq\left(\nu-\frac{\lambda^2}{4\gamma}\right)\left\|\Psi\right\|_2^2=\kappa\left\|\Psi\right\|_2^2. \tag{5.10}$$

This is the second inequality in (3.5). A nonzero critical point would make its left-hand side vanish, contradicting (5.10). Finally,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left\|\Psi(t)\right\|_2^2=-\langle D\mathcal{F}(\Psi(t)),\Psi(t)\rangle_{\mathbb{R}}\leq-\kappa\left\|\Psi(t)\right\|_2^2 \tag{5.11}$$

proves (3.6).

Remark 5.1 (meaning of the neutral result)**.** The result rejects nonzero equilibria only for the declared unconstrained, neutral, hash-pinned functional and its canonical gradient flow. It does not reject every retuned functional, fixed-charge problem, compact target, or conserved evolution. In particular, it is a candidate-functional rejection boundary, not a proof that a physical vacuum is empty.

6 Exact imposed-ensemble transition

6.1 The finite-torus spectral bottom

The quadratic operator has the form

$$\mathcal{L} = (r + Z(-\Delta) + Y(-\Delta)^2)\mathrm{I} + M, \quad M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \tag{6.1}$$

The characteristic polynomial of M is

$$p(t) = t^3 - \frac{2}{5}t^2 + \frac{223}{5000}t - \frac{3}{3125}. \tag{6.2}$$

An exact Sturm computation isolates its smallest root m_* by

$$\frac{2811617233511}{10^{14}} < m_* < \frac{2811617233512}{10^{14}}. \tag{6.3}$$

For $n \in \mathbb{Z}^3$, $|k_n|^2 = \pi^2|n|^2/64$. The rational enclosure of π used by the audit proves

$$\frac{5}{2} < \frac{-Z/(2Y)}{\pi^2/64} < \frac{7}{2}. \tag{6.4}$$

Because $|n|^2$ is integral and $3 = 1^2+1^2+1^2$, the radial minimum is the shell $|n|^2 = 3$, containing the eight body-diagonal indices $n = (\pm1, \pm1, \pm1)$. Thus

$$\lambda_0 = D(3\pi^2/64) + m_*, \qquad 0.28811617234458494031 < \lambda_0 < 0.28811617234459494032. \tag{6.5}$$

This is a finite-torus commensurability statement. It is not a BCC theorem and it does not remove the shell degeneracy in an infinite-volume limit.

6.2 Nonnegative decomposition and saturation

Set $Q = \|\Psi\|_2^2/2$. Coefficient matching for $f(\rho) = \lambda\rho^2/4 + \gamma\rho^3/6$ gives

$$\frac{\lambda_0}{2}\rho + f(\rho) = \frac{\mu_t}{2}\rho + \frac{\gamma}{6}\rho(\rho-\rho_*)^2, \quad \rho_* = -\frac{3\lambda}{4\gamma}, \quad c_* = \frac{3\lambda^2}{16\gamma}. \tag{6.6}$$

Adding the spectral and Class-II remainders yields (3.8).

Let u_* be a unit eigenvector of M for m_* and select any one of the eight indices on the shell. The plane wave

$$\Psi_*(x) = \sqrt{\rho_*}\, u_* \exp\left(\frac{2\pi i}{16} n \cdot x\right) \tag{6.7}$$

has constant ρ and constant internal bilinears. Hence $J_A = K_A = 0$, the spectral remainder vanishes, and

$$Q_* = \frac{16^3\rho_*}{2} = \frac{11008}{27}. \tag{6.8}$$

All terms in (3.8) are therefore saturated.

6.3 Fixed charge and grand potential

For a prescribed charge Q, let $\bar{\rho} = 2Q/16^3$. The exact Bregman identity is

$$f(\rho) - f(\bar{\rho}) - f'(\bar{\rho})(\rho - \bar{\rho}) = \frac{(\rho - \bar{\rho})^2}{12}(4\gamma\bar{\rho} + 2\gamma\rho + 3\lambda). \tag{6.9}$$

If $\bar{\rho} \geq \rho_*$, the bracket is nonnegative. On the affine constraint $\int \rho = 2Q$, the linear term integrates to zero. The spectral remainder, the Class-II energy, and (6.9) consequently prove constrained global minimality of a constant-density ground-shell plane wave for all $2Q/16^3 \geq \rho_*$. The constrained problem below Q_* is not decided here.

Subtracting μQ from (3.8) gives

$$\mathcal{O}_\mu(\Psi) = (\mu_t - \mu)Q(\Psi) + \frac{1}{2}\langle \Psi, (L - \lambda_0)\Psi\rangle_{\mathbb{R}} + \mathcal{F}_{\mathrm{II}}(\Psi) + \frac{\gamma}{6}\int \rho(\rho - \rho_*)^2 \,\mathrm{d}x. \tag{6.10}$$

For $\mu < \mu_t$, every nonzero field has positive grand potential. At $\mu = \mu_t$, zero and the saturated states coexist. For $\mu > \mu_t$, the plane wave has negative grand potential. We now make the existence step in this last assertion explicit. Write $\delta = \mu - \mu_t > 0$. Since the nonnegative Fourier symbol of $L - \lambda_0$ grows quadratically in $|k|^2$, there are constants $c > 0$ and $C_\delta < \infty$ such that

$$\frac{1}{2}\langle \Psi, (L - \lambda_0)\Psi\rangle_{\mathbb{R}} \geq c\,\|\Psi\|_{H^2}^2 - C_\delta\,\|\Psi\|_2^2\,. \tag{6.11}$$

The scalar polynomial also obeys, for $\rho \geq 0$,

$$\frac{\gamma}{6}\rho(\rho - \rho_*)^2 - \frac{\delta}{2}\rho \geq \frac{\gamma}{12}\rho^3 - C_\delta, \tag{6.12}$$

by Young's inequality. On the finite torus, the L^2 term in (6.11) is absorbed by the L^6 term in (6.12) up to a constant. Thus $\mathcal{O}_\mu$ is bounded below and coercive in H^2. If a minimizing sequence Ψ_n converges weakly in H^2, Rellich compactness gives strong convergence in H^1 and uniformly in space (after a subsequence). Hence the local potential is continuous, while the spectral quadratic form is weakly lower semicontinuous. Writing u_n and u for the realifications of Ψ_n and Ψ, the coefficients $B(u_n)$ in (4.7) converge uniformly to $B(u)$ and $\nabla u_n \rightharpoonup \nabla u$ in L^2. The positive quadratic Class-II integral is therefore weakly lower semicontinuous as well (equivalently, use the uniformly convergent positive square roots $B(u_n)^{1/2}$). The direct method therefore supplies a global minimizer; its value is negative because of the plane wave, so it is necessarily nonzero.

The zero-field Hessian loses positivity at $\mu_{\mathrm{lin}} = \lambda_0$. Along the ground-plane-wave amplitude family, stationary densities solve

$$\gamma\rho^2 + \lambda\rho + (\lambda_0 - \mu) = 0. \tag{6.13}$$

The amplitude saddle-node is $\mu_{\mathrm{sn}} = \lambda_0 - \lambda^2/(4\gamma)$, while coexistence is $\mu_t = \lambda_0 - 3\lambda^2/(16\gamma)$. With the pinned coefficients this is exactly (3.10); the global charge jumps from zero to Q_* while zero remains a strict local minimum. This is the precise sense in which the imposed ensemble transition is first order.

Remark 6.1 (why R-157 and R-158 do not conflict)**.** At the nonzero state (6.7), the constrained Euler–Lagrange equation is $D\mathcal{F}(\Psi_*) = \mu_t\Psi_*$, not $D\mathcal{F}(\Psi_*) = 0$. R-157 concerns the latter equation in the unconstrained linear space; R-158 changes the variational problem. Moreover, $\mathcal{F}(\Psi_*) = \mu_t Q_* > 0 = \mathcal{F}(0)$, so the ordered state does not beat the original neutral reference.

7 Relation to prior work and residual contribution

The functional-analytic ingredients used here are standard: sectorial fourth-order semigroups, Sobolev product estimates, Galerkin compactness, weakly singular Gronwall inequalities, and polynomial coercivity. A Swift–Hohenberg analysis by Giorgini [3] proves well-posedness and long-time attractor properties for scalar slow/fast equations, but does not contain the present three-component regularized Class-II map, its exact radial sign, or the imposed-ensemble completion. Thus the well-posedness theorem alone is not claimed as a new general method.

Quantum anharmonic-crystal phase-transition work, such as Kargol–Kondratiev–Kozitsky [4] develops Euclidean Gibbs states, reflection arguments, and infrared estimates for specified lattice models. The present paper does not import a quantum phase theorem. It instead treats a classical fixed-torus variational functional and proves the model-specific sign and spectral identities needed for its two ensembles. The residual contribution is therefore the exact combination of (a) the full derivative Class-II well-posedness argument, (b) the global neutral rejection, and (c) the finite shell ensemble classification, all with a common pinned functional and an explicit scope firewall.

The claim is not a world-first assertion. A complete novelty determination requires a specialist literature search beyond the bounded crosswalk recorded in `literature-crosswalk.md` and independent mathematical review of the model-specific estimates.

8 Verification and reproducibility

The paper cites only the registered A2 claim and its result records:

component	independent executable evidence	recorded result
full evolution	four audits	20/20, 14/14, 12/12, 15/15 PASS
R-157 neutral sign	primary and non-importing lanes	26/26, 24/24 PASS
R-158 ensemble sign	primary and standard-library lanes	35/35, 24/24 PASS
R-472 exact core	Lean sidecar and hostile checks	30/30, 22/22, 12/12, 22/22 PASS

The canonical commands are listed in `verification/README.md`. The tests recompute rational constants, matrix signs, spectral intervals, normalizations, and hostile mutations. They do not prove the analytic semigroup, compactness, chain-rule, or external-literature steps; those remain the proof text's responsibility.

The 2026-09-03 replay passed the A2 wrapper and all four R-157/R-158 child lanes. After the structured T-054 predicate repair recorded in EXP-001372, the R-157 integrated lane passes 144/144 (legacy A2 61/61) and the R-158 integrated lane passes 155/155 with the R-157/A2 regression. This closes the repository synchronization gate only; the paper remains at draft pending the analytic proof audit, specialist literature review, operator confirmation, capstone bundle, and release checks.

The existing A2 T6 reproduction bundle is recorded in the claim card and its bundle directory. A dedicated capstone bundle containing the integrated R-157/R-158 referee note and all transitive scripts is a required post-review action. Under the repository policy it must be built only after operator confirmation of the integrated referee package. This draft does not claim that confirmation or a release-gated submission.

9 Limitations and falsifiers

The theorem's domain is the fixed side-16 torus, six-real-component H^2 field space, pinned coefficients, positive density floor, and $\eta_{\text{shell}} = 0$. The following are outside its scope:

- infinite-volume or continuum removal, counterterm flow, and regulator independence;

- a real-time infinite-volume dynamics, algebraic KMS states, ground states, spectral gaps, extremality, purity, or clustering;
- a conserved physical charge, a reservoir, or a derivation of μ;
- gauge completion, gauge-invariant density modulation, BCC or unique morphology;
- the historical non-variational backend, signed stochastic composites, alternative parameters, and compact target spaces;
- any physical-vacuum, gravity, cosmology, or Sector-A closure claim.

The analytic theorem is falsified by a counterexample to global H^2 well-posedness, uniqueness, continuous dependence, the exact energy identity, or positive-time smoothing in the declared domain. R-157 is falsified by an exact nonzero critical point or a field violating (3.5). R-158 is falsified by failure of the exact spectral shell, the decomposition, the Bregman sign, plane-wave saturation, or the ordering (3.10). A failure of the named P1 identification hypothesis blocks transfer to the TECT production interpretation but does not alter the mathematics of the explicitly defined functional.

10 Conclusion

The pinned functional has a sharp two-sided mathematical verdict. Its neutral unconstrained dynamics are globally well posed and converge exponentially to zero; no nonzero equilibrium survives the exact radial sign. A nonzero shell state appears only after the variational problem is changed by an imposed charge or chemical potential, where an exact first-order coexistence mechanism can be proved on the finite torus. This separation is the principal result: the ensemble mechanism is mathematically real, while its physical provenance and interpretation remain open. The manuscript is therefore a candidate independent mathematical paper, currently at draft status pending the literature crosswalk, proof audit, operator confirmation, capstone bundle, and release checks specified in the accompanying files.

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